Validation results

All 43 checks passed in real Logisim-Evolution 5.0.0 simulation with OpenJDK 21.0.2. Every recorded test had zero error-valued endpoints. The six required groups and additional boundary tests are reproducible with run_tests.bat.

1. Data hazards

Initial inputs and actions: Reset; R1=5, R2=3, R3=20 for forwarding. Load-use stores 7 at RAM 10 hex. Youngest test starts R1=10, R2=3, R3=4 and address R4=12 hex.

Expected: Forwarding gives R1=8, R3=12 with no stall. Load-use gives R3=12 after exactly one stall. Youngest store writes 9 once.

Observed: Observed exactly those register values, one load-use bubble and one store per intended ST. PC and IF/ID held during the bubble; older stages advanced.

2. Control transfer

Initial inputs and actions: Reset; SUB R1,R1 makes zero before BEQZ. Put ST to valve and ADD on the wrong path. A separate branch puts HALT on the wrong path.

Expected: Neither wrong-path instruction may commit. Target code must still run despite wrong-path HALT.

Observed: No wrong-path stores occurred. Target R3=85; wrong-path HALT test reached R2=85. Valid HALT drained older writes and held fetch until reset.

Common setup: active reset for one full cycle, then stable sensor inputs around MEM sampling edges. CSV values are hex except cycle and errors. Stimulus files preserve every input change and clock count.

3. Memory isolation

Initial inputs and actions: Reset; write AB hex to RAM 22 hex. Attempt writes at reserved 80 and EF, then to read-only water F1 with water input 77.

Expected: RAM reads AB; reserved reads zero; water remains 77. Invalid stores have no effect.

Observed: Observed R3=AB, R4=00, R5=00, R6=4D hex. All 128 RAM and 112 reserved addresses also passed; invalid RAM/I/O writes changed nothing.

4. Timer

Initial inputs and actions: Reset count/match. Enable with limit 3, clear and rearm with limit 5. Test wrap, zero limit, disabled hold and CPU stall/HALT.

Expected: MATCH sets on increment to limit; stays set through wrap; clear wins. Timer continues through stalls/HALT.

Observed: Count 3 had MATCH=1; clear gave count/match zero; after 260 increments count=4 and MATCH=1. Zero limit did not match. MMIO intervals, edge clear and independent ticking passed.

5. Safety interlock

Initial inputs and actions: Reset; door closed, request on. Open door without a clock, then sample one rising edge. Reclose without rearm. Try unsafe START edges, then request off and a fresh closed-door START edge.

Expected: Physical disable is immediate. TRIP sets only on edge and persists after closing. Only the safe fresh edge clears it.

Observed: Actual output changed 1 to 0 before the edge while TRIP remained 0. Next edge latched TRIP=1; closing did not restart. Unsafe rearm failed; safe rearm passed.

6. Complete wash

Initial inputs and actions: Reset, closed door and START accepted. Mode 0 water 99→102→9; mode 1 water 149→153→9. Change MODE after acceptance. Separate test uses exact thresholds 100 and 10.

Expected: Snapshot parameters; stop filling after overshoot; both motor directions; drain; spin; all actuators off and unlocked, DISPLAY=01, BUZZER=1.

Observed: Both modes completed. Timer limits written were 5,5,12 and 8,8,16. Both directions appeared. Final requests and lock were zero, display 01 and buzzer 1. Exact thresholds and the real main-panel clock also passed.